



Mohit Kumar

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B.Tech in Mechatronics and Automation Engineering

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Portfolio

EDUCATION

Indian Institute Of Information Technology, Bhagalpur

2025 – 2029

B.Tech in Mechatronics and Automation Engineering

CGPA: 8.02

TECHNICAL SKILLS AND INTERESTS

Languages: C, C++, C#, Python, SQL

Tools & Platforms: ESP32-S3, Arduino, Raspberry Pi, ESP-IDF, PlatformIO, Unity, Fusion 360, Ansys, MATLAB, SolidWorks, Blender, GitHub, VS Code

Areas of Interest: Robotics, Embedded Systems, Pneumatic Artificial Muscles, IoT Automation, Game Development, Competitive Programming

EXPERIENCE

Research Intern – IIT Guwahati

May 2026 – July 2026

Pneumatic Exoskeleton System (PAM-based)

Robotics And Mechatronics Lab

- Designed and modelled the Compressed Energy Unit (CEU) – housing solenoid valves, pressure regulators, and manifold – in Fusion 360; performed structural simulation in Ansys and validated pneumatic circuit behaviour in MATLAB
- Developed an observer-based feedback controller with predictive PID compensation to offset PAM actuation latency; implemented FreeRTOS balance modulation logic on ESP32-S3 with BNO055 IMU-based state estimation

PROJECTS

Offline Voice Recognition System

Mar 2026 – Present

Offline voice command pipeline on ESP32-S3 for smart home automation; master-slave architecture, no cloud dependency

- Stack: ESP32-S3, ESP-IDF, ESP-SR (WakeNet wn9s_hiesp + MultiNet), PlatformIO, MQTT
- Master node handles I2S mic capture, energy-gated wake detection, and phoneme-based command classification; slave nodes execute home automation actions over MQTT

Line Following Robot (LFR) & 3D Maze Solver

Feb 2026

Autonomous robots using sensor arrays and pathfinding algorithms; 3rd Prize at Enyugma, IIIT Bhagalpur

- Stack: Arduino, C++, IR sensor array, L298N motor driver; tuned PID control loop for stable high-speed line tracking with smooth cornering
- Engineered maze-solving logic for autonomous 3D navigation using wall-following and decision-node mapping; integrated multi-sensor fusion for reliable path detection

2D Action Platformer (GDAI Game Jam)

Feb 2026

Solo-developed 2D action platformer in 40 hours at Enyugma; won 3rd Prize in Game Dev event

- Stack: Unity, C#; architected core gameplay loop with state-machine-driven enemy AI, multi-hit combo system, frame-perfect animations, and progressive level design
- Contributed ~80% of the project individually under jam constraints; handled game feel tuning, collision logic, and scene management end-to-end

POSITIONS OF RESPONSIBILITY

Lead – Vector, Game Dev Club

Jan 2026 – Present

Position of Responsibility

IIIT Bhagalpur

- Leading game dev initiatives and member coordination; organised Enyugma game dev event

Core Member – Robotics Club REC

Jan 2026 – Present

Position of Responsibility

IIIT Bhagalpur

- Contributed to embedded robotics projects; represented club at Enyugma techno-cultural fest

ACHIEVEMENTS

CodeChef 2-Star | 450+ Problems Solved

2025 – Present

Active competitive programmer; consistent participant in rated CodeChef and Codeforces contests

3rd Prize – Robotics & Game Dev Events, Enyugma

Feb 2026

Won both Robotics (LFR + Maze Solver) and Game Development events at IIIT Bhagalpur's Techno-Cultural Fest